

Software Requirements Specification (SRS)

AI-Based Employee Stress Management & Smart Workload System

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Document Title: Software Requirements Specification (SRS)

Project Title: AI-Based Smart Office Employee Stress Detection, Monitoring and Workload Management System

Project Type: Software Development Oriented Project

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1. Introduction

1.1 Purpose

The purpose of the **Stress Level Smart Office System** is to monitor, analyze, and manage employee stress levels in a smart office environment using artificial intelligence and real-time data. The system aims to identify employee stress through facial expression analysis and provide timely insights to employers and administrators in order to improve employee well-being and workplace productivity.

This system is designed to assist organizations in making informed decisions related to workload allocation, task scheduling, and employee support by continuously evaluating stress levels during working hours. Based on the detected stress levels, the system helps employers adjust workloads, reduce excessive pressure, and maintain a healthy working environment.

Additionally, the system seeks to minimize human bias in stress assessment by using AI-based automated detection techniques, ensuring objective and consistent results. By integrating stress monitoring with task management and reporting features, the Stress Level Smart Office System supports proactive stress management, improves employee performance, and contributes to overall organizational efficiency.

1.2 Scope

Stress level is a web-based smart office platform that detects employee stress using AI-based facial expression analysis and dynamically manages workloads based on detected stress levels. The system supports real-time stress checks, task distribution, notifications, and analytics dashboards.

1.3 Definitions, Acronyms, and Abbreviations

Term	Description
SRS	Software Requirements Specification
AI	Artificial Intelligence
Admin	System administrator
Employer	Manager who assigns workloads
Employee	End user whose stress is monitored
API	Application Programming Interface
JWT	JSON Web Token

1.4 References

Lee, K. S., & Park, J., “A Survey on Stress Detection Using Facial Expression Analysis,” *IEEE Access*, 2020.

Khan, M. A., Zhang, Y., Sharif, M., & Akram, T., “A Facial Expression Recognition Framework Using Deep Learning for Stress Detection,” *IEEE Access*, 2021.

Saha, S., Datta, S., & Konar, A., “Real-Time Stress Detection Using Facial Expressions and Convolutional Neural Networks,” *Proceedings of IEEE International Conference on Artificial Intelligence and Signal Processing (AISP)*, 2021.

Alsharif, M. H., Kim, S., & Kuruoğlu, E., “Facial Emotion-Based Stress Detection Using Deep Neural Networks,” *IEEE Transactions on Affective Computing*, 2022.

Rahman, M. M., Hossain, M. S., & Muhammad, G., “Multimodal Stress Detection Using Facial Expressions and Deep Learning,” *IEEE Access*, 2022.

Patel, D., Shah, H., & Patel, R., “Mental Stress Level Classification Using Facial Expression Analysis and Machine Learning,” *Proceedings of IEEE International Conference on Smart Computing and Systems Engineering (SCSE)*, 2023.

Zhou, Y., Wang, X., & Liu, J., “Vision-Based Stress Detection Using Facial Micro-Expressions and Deep Learning,” *IEEE Sensors Journal*, 2023.

1.5 Overview

The Stress Level Smart Office System is a web-based application designed to support intelligent stress monitoring and workload management within an office environment. The system is composed of four main components: a frontend web application, a FastAPI-based backend server, a machine learning-based stress detection module, and a centralized database for data storage and management.

The frontend web application provides interactive user interfaces for different user roles, including Admin, Employer, and Employee. It allows users to securely access the system, view dashboards, manage tasks, request or perform stress checks, and receive notifications. The backend, developed using FastAPI, handles business logic, authentication, authorization, data processing, and communication between the frontend, database, and AI modules through RESTful APIs.

The machine learning stress detection module analyzes facial expressions captured through a camera to identify stress levels using trained models. Detected stress data, along with task and user information, is securely stored in the database for real-time access, historical analysis, and report generation.

Role-based access control ensures that each user interacts only with the features relevant to their responsibilities. Administrators manage users and system configurations, employers monitor employee stress levels and assign workloads, and employees view their tasks, stress status, and notifications. Overall, the system integrates artificial intelligence with smart office management to promote employee well-being, improve productivity, and support data-driven decision-making.

2. Overall Description

2.1 Product Perspective

The Stress Level Smart Office System is developed as a web-based application following a client–server architecture. In this architecture, the frontend acts as the client and is accessed through standard web browsers running on Windows-based systems, while the backend services and artificial intelligence components are deployed on an Ubuntu server environment.

The frontend is responsible for user interaction, data visualization, and input collection, providing role-specific interfaces for Admin, Employer, and Employee users. All client requests are transmitted securely to the backend server, which manages authentication, authorization, business logic, and data processing.

The backend, implemented using the FastAPI framework, serves as the core processing unit of the system. It communicates with the machine learning–based stress detection module, which analyzes facial expressions captured via camera devices to determine employee stress levels. The backend also interacts with the database to store and retrieve user information, stress records, task details, and notification data.

Additionally, the system integrates multiple functional services, including task management, stress analytics, reporting, and notification handling, into a unified platform. This modular design allows each component to operate independently while ensuring seamless communication through RESTful APIs, making the system scalable, maintainable, and adaptable to future enhancements.

2.1.1 System Interfaces

The Stress Level Smart Office System provides system interfaces that enable communication between users, system components, and external services. The primary system interface is a web browser interface, which allows users to access the system through standard web browsers without requiring additional software installation. This interface supports interaction with the system’s frontend, including user authentication, dashboard visualization, task management, stress monitoring, and notification handling.

The system also exposes RESTful backend APIs implemented using the FastAPI framework. These APIs serve as the communication layer between the frontend application, the machine learning stress detection module, and the database. All requests and responses are exchanged in JSON format over secure HTTP/HTTPS protocols. The RESTful APIs handle operations such as user management, task assignment, stress data processing, report generation, and notification delivery.

This interface-based architecture ensures loose coupling between system components, enabling scalability, easier maintenance, and future integration with additional services or platforms if required.

2.1.2 User Interfaces

The Stress Level Smart Office System provides multiple user interfaces designed to support different user roles and ensure ease of use, clarity, and efficiency. All user interfaces are web-based and accessible through standard web browsers.

The **Login and Registration pages** allow users to securely create accounts and authenticate themselves into the system. During registration, users can select their role, and during login, credentials are verified through the backend authentication mechanism.

The **Admin Dashboard** provides a comprehensive system overview and is designed for system administrators. This interface enables the admin to manage users, monitor overall system activity, view summarized stress and workload statistics, and ensure the smooth operation of the system.

The **Employer Dashboard** is designed for managers or supervisors. It allows employers to request stress checks for employees, view stress analysis results, manage workloads, assign or adjust tasks based on stress levels, and monitor employee performance trends.

The **Employee Dashboard** allows employees to view their current stress status, accept or decline stress check requests, manage assigned workloads, and receive system notifications. This interface promotes transparency by enabling employees to be aware of their stress levels and assigned responsibilities.

All user interfaces are designed to be responsive, intuitive, and role-specific, ensuring that users can efficiently interact with the system with minimal training.

2.1.3 Hardware Interfaces

The primary hardware interface required by the Stress Level Smart Office System is a **webcam**, which is used for real-time facial expression capture during stress detection processes. The webcam enables the system to obtain live facial images of employees, which are then analyzed by the machine learning-based stress detection module.

The webcam can be either an integrated camera (commonly available in laptops) or an external USB camera connected to a desktop computer. The system accesses the webcam through the web browser after obtaining user consent, ensuring privacy and security compliance.

Captured facial data is processed temporarily for stress analysis and is not stored as raw video unless explicitly required for system improvement or model training. The hardware interface is designed to work with standard, low-cost camera devices, making the system practical and easily deployable in typical office environments.

2.1.4 Software Interfaces

The Stress Level Smart Office System interacts with several software components that together enable its core functionalities. The primary software interface is the **FastAPI backend**, which acts as the central processing unit of the system. The FastAPI backend handles user authentication, role-based access control, business logic execution, API request handling, and coordination between the frontend, database, and machine learning modules.

The system also interfaces with an **OpenCV-based stress detection module**, which is responsible for processing facial images captured through the webcam. This module performs facial feature extraction and applies trained machine learning or deep learning models to analyze facial expressions and determine the stress level of employees. The stress detection results are then sent back to the backend through defined interfaces for further processing and storage.

Additionally, the system interfaces with a **MongoDB database**, which is used to store user profiles, stress detection records, workload details, task assignments, notifications, and system logs. MongoDB provides flexible and scalable data storage, supporting efficient handling of both structured and semi-structured data generated by the system.

These software interfaces communicate through well-defined APIs and data formats, ensuring modularity, maintainability, and ease of integration for future enhancements.

2.1.5 Communication Interfaces

The Stress Level Smart Office System uses secure and standardized communication interfaces to enable reliable data exchange between system components and users. All communication between the frontend web application and the backend server is conducted using the **HTTPS protocol**, which ensures data confidentiality, integrity, and protection against unauthorized access during transmission.

The system follows a **JSON-based RESTful API** communication model for all client-server interactions. Requests and responses exchanged between the frontend, backend, machine learning module, and database services are formatted in JSON, providing a lightweight, human-readable, and platform-independent data exchange mechanism.

RESTful APIs are used to handle operations such as user authentication, stress detection requests, task and workload management, notification delivery, and report generation. This communication approach ensures interoperability, scalability, and ease of integration with additional services or future system extensions.

2.1.6 Memory Constraints

The Stress Level Smart Office System requires adequate memory resources to ensure smooth operation, particularly for the AI-based stress detection services. A minimum of **8GB RAM** is recommended for the backend server hosting the FastAPI services, machine learning modules, and database operations.

The memory requirement is influenced by real-time facial image processing, machine learning model execution, concurrent user requests, and data storage operations. Sufficient memory allocation helps maintain acceptable system response times and ensures accurate stress detection without performance degradation.

In environments with higher user loads or advanced deep learning models, additional memory resources may be required to support scalability and future system enhancements.

2.1.7 Operations

The Stress Level Smart Office System supports several core operational processes that enable effective stress monitoring and workload management within the organization. These operations are executed through coordinated interactions between users, system components, and AI services.

The system provides **user authentication and role management**, allowing users to securely log in and access system features based on their assigned roles, such as Admin, Employer, or Employee. Role-based access control ensures that each user can perform only authorized operations.

The system supports a **stress check request and approval workflow**, where employers can request stress assessments for employees, and employees can accept or decline these requests. Upon approval, the system initiates the stress detection process.

The **camera-based stress detection operation** captures facial expressions through a webcam and analyzes them using the AI-based stress detection module to determine the employee's stress level. The detected results are processed and stored for further analysis and reporting.

The system also handles **workload creation and auto-assignment**, where tasks or workloads are created by employers and automatically assigned or adjusted based on detected stress levels. This operation helps balance employee workload and reduce excessive stress.

Additionally, the system manages **notifications and reporting**, providing real-time alerts to users regarding stress requests, results, and workload changes, as well as generating reports for monitoring trends and decision-making.

2.1.8 Site Adaptation Requirements

The Stress Level Smart Office System is designed to operate in a distributed environment with specific site adaptation requirements. The backend components, including the FastAPI server, AI-based stress detection module, and database services, shall be deployed on an **Ubuntu server** to ensure stability, security, and efficient AI processing.

End users access the system through **Windows-based client systems** using standard web browsers. The system does not require additional client-side installations, allowing easy deployment and accessibility. Proper network connectivity between the Ubuntu server and Windows clients is required to ensure secure and reliable system operation.

2.2 Product Functions

The Stress Level Smart Office System provides a comprehensive set of functions designed to support intelligent stress monitoring and workload management in an office environment.

The system offers **secure login and signup functionality with role selection**, allowing users to register and authenticate as Admin, Employer, or Employee. Role-based access control ensures that users can access only the features relevant to their assigned roles.

The system enables **admin user management**, allowing administrators to create, update, and manage user accounts, monitor system activity, and maintain overall system integrity.

Employers can initiate **stress check requests** for employees through the system. Employees have the ability to **approve or decline stress check requests**, ensuring user consent and transparency in stress monitoring activities.

The system performs **AI-based stress level calculation**, analyzing facial expressions captured through a webcam to classify stress levels as Low, Medium, or High. Along with the classification, the system calculates a confidence level to indicate the reliability of the detected result.

The system maintains **stress history tracking**, storing past stress records with timestamps and confidence values to support trend analysis and reporting.

Based on detected stress levels, the system supports **smart workload distribution**, automatically assigning or adjusting workloads to help balance employee pressure and improve productivity.

Additionally, the system provides **real-time notifications** to inform users about stress requests, results, workload updates, and important system events, ensuring timely communication and responsiveness.

2.3 User Characteristics

The Stress Level Smart Office System is designed to be used by different categories of users, each with distinct roles and responsibilities.

The **Admin** user is expected to have technical knowledge and experience in system supervision. Admin users are responsible for managing user accounts, configuring system settings, monitoring system performance, and ensuring the overall stability and security of the system.

The **Employer** user is typically a manager or supervisor with strong managerial skills and basic information technology knowledge. Employers use the system to monitor employee stress levels, initiate stress check requests, manage workloads, review reports, and make informed decisions to improve workplace efficiency and employee well-being.

The **Employee** user is expected to have basic computer usage skills and minimal technical knowledge. Employees interact with the system to log in, respond to stress check requests, view their stress status, manage assigned workloads, and receive notifications. The user interfaces are designed to be intuitive, enabling employees to use the system with little or no prior training.

2.4 Constraints

The Stress Level Smart Office System is subject to several constraints that may affect its design, implementation, and operation.

The system requires that the **stress detection accuracy be at least 70%** to ensure that the results provided by the AI-based stress detection module are sufficiently reliable for decision-making. This constraint influences the selection of datasets, machine learning models, and training techniques used in the system.

The system depends on **webcam access and user permission** to perform facial expression-based stress detection. Without webcam availability or user consent, the stress detection functionality cannot be executed.

Additionally, **internet connectivity is required** for system operation, as the frontend web application must communicate with the backend server, AI services, and database through secure network connections. Limited or unstable internet access may affect system performance and availability.

These constraints must be considered during deployment and usage to ensure proper system functionality.

2.5 Assumptions and Dependencies

The Stress Level Smart Office System is developed based on several assumptions and external dependencies that are necessary for its proper operation.

It is assumed that all users provide **explicit consent for camera usage** when participating in stress detection activities. The system relies on this consent to capture facial expressions through the webcam and to perform AI-based stress analysis in a compliant and ethical manner.

The system also depends on the availability of an **adequate and representative training dataset** for the machine learning stress detection module. The accuracy and reliability of stress detection are directly influenced by the quality, size, and diversity of the training data used to develop the model.

Additionally, the system assumes the availability of required hardware resources, stable internet connectivity, and a properly configured server environment. Any changes to these assumptions or dependencies may impact system performance and detection accuracy.

2.6 Apportioning of Requirements

The requirements of the Stress Level Smart Office System are planned to be implemented in a phased manner to ensure systematic development and stability. In the initial phase, core system functionalities such as **user management, authentication, role-based access control, and basic task and workload handling** will be implemented and thoroughly tested.

Once the foundational features are stable, the next phase will focus on integrating the **AI-based stress detection module**, including facial expression capture, stress classification, and result storage. Advanced features such as **stress optimization techniques, workload auto-adjustment based on stress trends, and performance enhancements** will be implemented in later stages.

This phased approach allows early validation of essential system functions while providing flexibility for improving AI accuracy, optimizing performance, and incorporating future enhancements without disrupting core operations.

3. Specific Requirements (SRS)

3.1 External Interface Requirements

The Stress Level Smart Office System shall provide well-defined external interfaces to ensure effective interaction between users and system components. The system shall include a **responsive web-based user interface (UI)** that adapts to different screen sizes and resolutions, allowing users to access the system from various devices using standard web browsers. The UI shall support smooth navigation, clear visualization of data, and role-specific functionalities for Admin, Employer, and Employee users.

In addition, the system shall expose **secure REST API endpoints** to handle communication between the frontend application, backend services, and the AI-based stress detection module. These APIs shall follow RESTful principles and exchange data in JSON format over HTTPS. Security mechanisms such as authentication tokens and role-based authorization shall be enforced to prevent unauthorized access to system resources.

These external interfaces ensure interoperability, scalability, and secure communication, enabling reliable system operation and future integration with additional services if required.

3.2 Functional Requirements

The functional requirements define the core functionalities that the Stress Level Smart Office System must provide in order to meet user and system objectives.

FR-01: The system shall allow users to register and log in with role selection (Admin, Employer, or Employee) using secure authentication mechanisms.

FR-02: The system shall allow the employer to initiate a stress check request for a selected employee through the employer dashboard.

FR-03: The system shall allow the employee to accept or decline a stress check request before the stress detection process is initiated.

FR-04: The system shall capture facial images in real time using a webcam after obtaining user consent for stress analysis.

FR-05: The system shall analyze captured facial expressions and classify the employee's stress level as **Low**, **Medium**, or **High** using an AI-based model.

FR-06: The system shall calculate and display the stress percentage and confidence level associated with each stress detection result.

FR-07: The system shall store stress detection history, including timestamps, stress level, and confidence values, for future analysis and reporting.

FR-08: The system shall automatically assign or adjust workloads based on the detected stress level to help balance employee workload and reduce excessive stress.

FR-09: The system shall notify relevant users about stress check requests, stress detection results, and workload updates through real-time notifications.

3.3 Classes / Objects

Class	Description
User	Authentication and role data
StressRecord	Stress results and confidence
Workload	Task details and category
Notification	User alerts

3.4 Performance Requirements

The Stress Level Smart Office System shall meet specific performance requirements to ensure a responsive and efficient user experience. The **API response time** for all standard system operations, including user authentication, stress check requests, data retrieval, and workload updates, shall be **less than 3 seconds** under normal operating conditions.

The **stress detection latency**, which includes facial image capture, processing, and stress level classification by the AI module, shall be **less than 5 seconds** per stress detection session. This ensures that users receive timely feedback without noticeable delays.

These performance requirements are critical for maintaining real-time interaction, supporting multiple concurrent users, and ensuring smooth integration between the frontend, backend, and AI-based stress detection services.

3.5 Design Constraints

The design and implementation of the Stress Level Smart Office System are subject to specific constraints that guide technology selection and system architecture.

The backend of the system **must be developed using the FastAPI framework**. This constraint ensures high performance, scalability, and efficient handling of RESTful API requests, as well as seamless integration with Python-based AI modules.

The **AI-based stress detection model must be implemented using Python and OpenCV**. This constraint is imposed to support facial expression processing, real-time image analysis, and compatibility with existing machine learning libraries. OpenCV provides essential computer vision functionalities required for facial feature extraction and stress analysis.

These design constraints ensure consistency in development, simplify integration between system components, and support maintainability and future enhancements of the system.

3.6 Software System Attributes

The Stress Level Smart Office System is designed to meet the following software quality attributes to ensure reliable and secure operation.

Reliability:

The system shall maintain a minimum of **95% uptime**, ensuring consistent and dependable operation during normal working hours. Proper error handling, logging, and recovery mechanisms shall be implemented to minimize system failures and ensure stable performance.

Availability:

The system shall be available **24 hours a day, 7 days a week**, allowing users to access system features such as stress monitoring, workload management, and reporting at any time, subject to scheduled maintenance periods.

Security:

The system shall implement **JWT-based authentication** to securely verify user identities. **Role-based access control (RBAC)** shall be enforced to restrict system functionalities according to user roles (Admin, Employer, Employee), ensuring data confidentiality and preventing unauthorized access.

Maintainability:

The system shall follow a **modular and scalable architecture**, enabling easy maintenance, debugging, and future enhancements. Clear separation of components such as frontend, backend, AI modules, and database services allows the system to be updated or extended with minimal impact on existing functionalities.

4. Supporting Information

4.1 Appendices

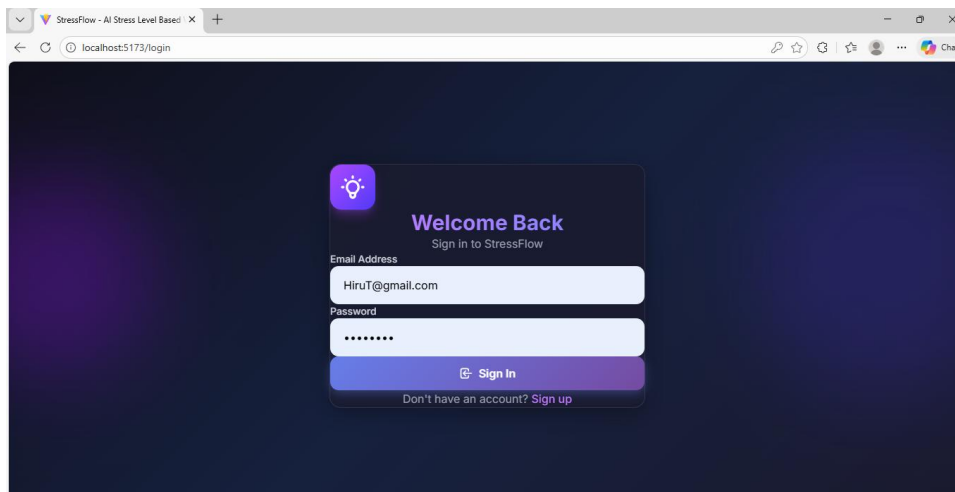
The appendices provide additional supporting details that enhance the understanding of the Stress Level Smart Office System. These sections include visual evidence, technical summaries, and logical explanations that complement the main requirements described in this document.

Appendix A – User Interface Screenshots

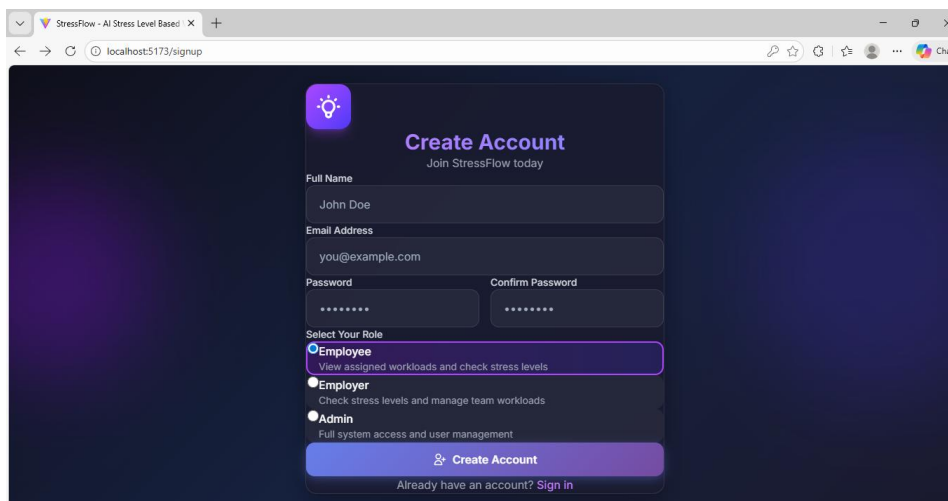
This appendix contains screenshots of the key user interfaces of the system. These screenshots are provided to illustrate the design, layout, and functionality of the user interfaces implemented in the system.

The included screenshots represent:

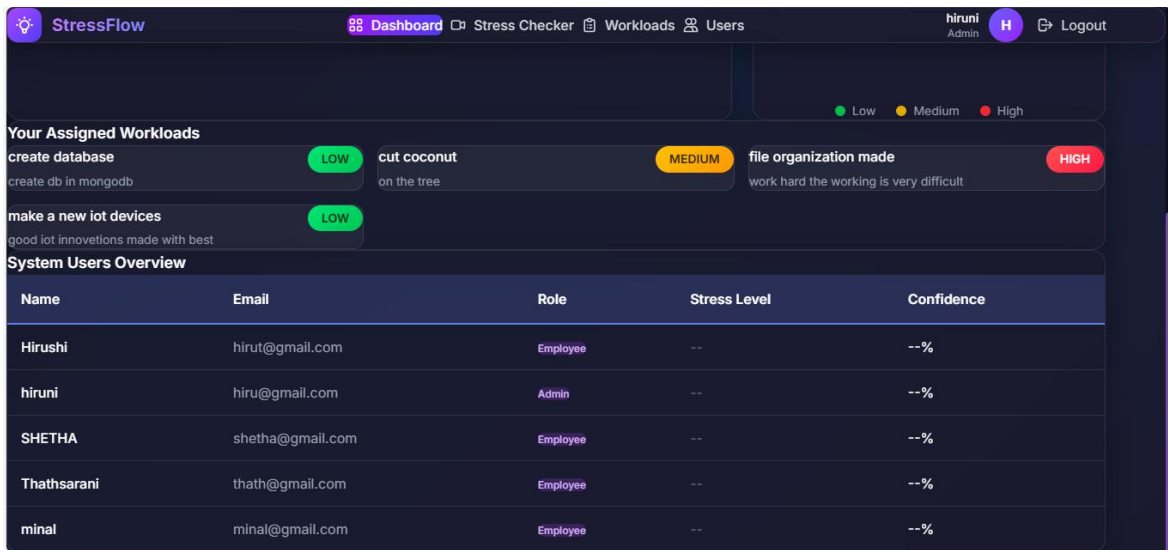
- Login page



- Signup / Registration page



- Admin dashboard showing user and system overview



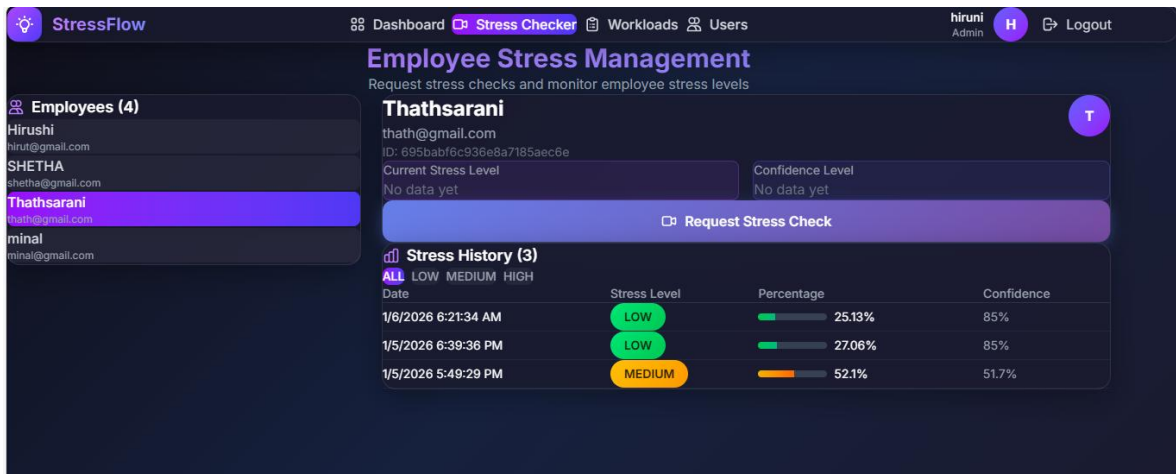
The dashboard shows a top navigation bar with 'StressFlow' logo, a menu (Dashboard, Stress Checker, Workloads, Users), and user info (hiruni Admin, Logout). A legend indicates Low (green), Medium (yellow), and High (red) stress levels.

Your Assigned Workloads

- create database (LOW): create db in mongodb
- cut coconut (MEDIUM): on the tree
- file organization made (HIGH): work hard the working is very difficult
- make a new iot devices (LOW): good iot innovations made with best

System Users Overview

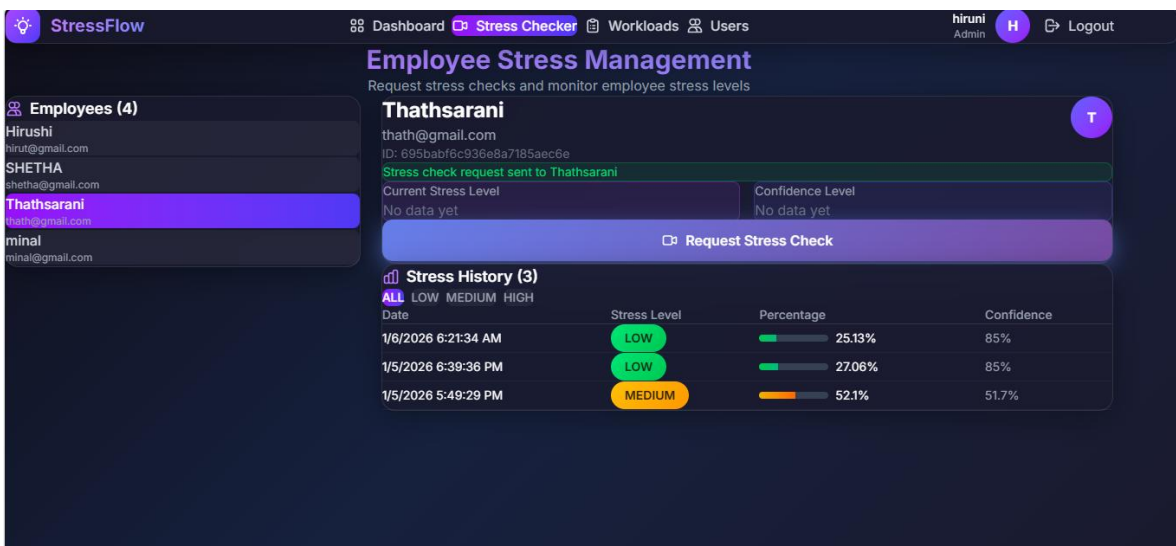
Name	Email	Role	Stress Level	Confidence
Hirushi	hirut@gmail.com	Employee	--	--%
hiruni	hiru@gmail.com	Admin	--	--%
SHETHA	shetha@gmail.com	Employee	--	--%
Thathsarani	thath@gmail.com	Employee	--	--%
minal	minal@gmail.com	Employee	--	--%



The 'Employee Stress Management' section for Thathsarani (thath@gmail.com, ID: 695babf6c936e8a7185aec6e) shows a 'Request Stress Check' button and a 'Stress History' table.

Stress History (3)

Date	Stress Level	Percentage	Confidence
1/6/2026 6:21:34 AM	LOW	25.13%	85%
1/5/2026 6:39:36 PM	LOW	27.06%	85%
1/5/2026 5:49:29 PM	MEDIUM	52.1%	51.7%

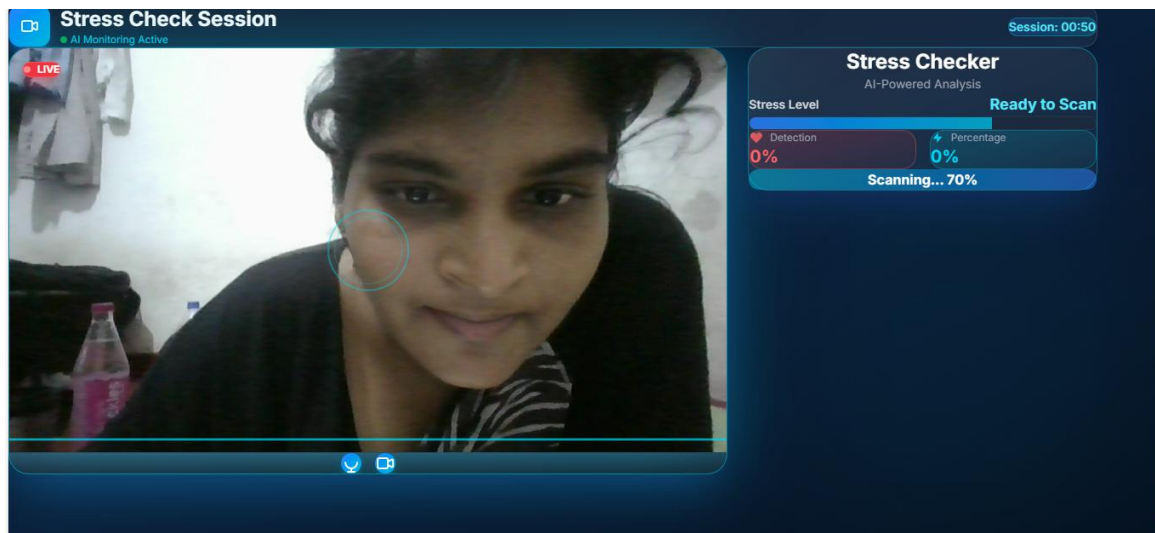
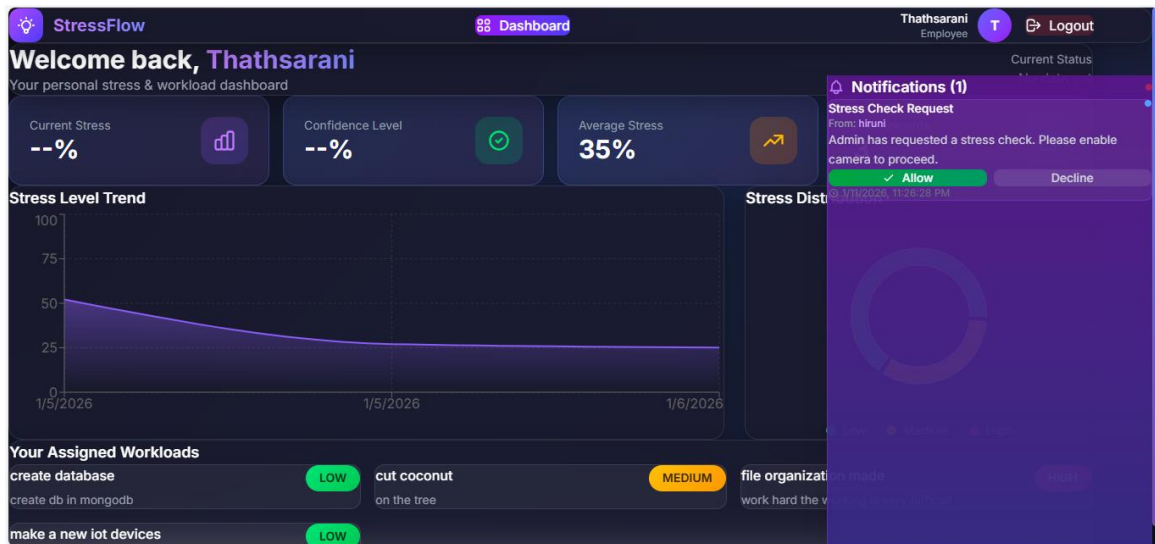


This view is identical to the previous one but includes a green notification banner: 'Stress check request sent to Thathsarani'.

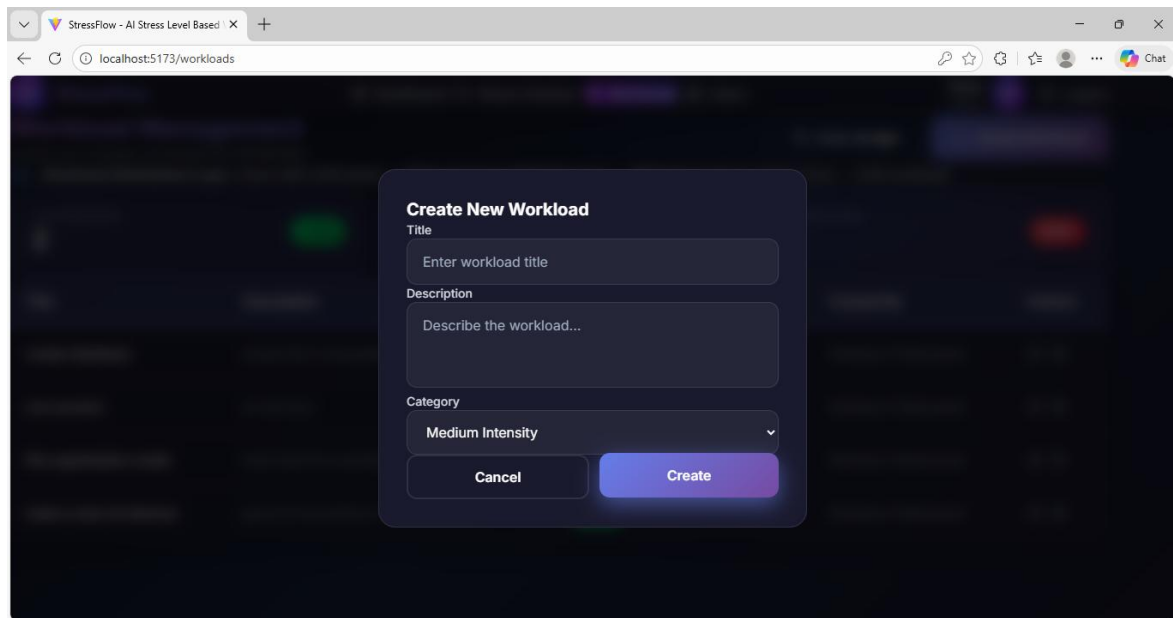
Stress History (3)

Date	Stress Level	Percentage	Confidence
1/6/2026 6:21:34 AM	LOW	25.13%	85%
1/5/2026 6:39:36 PM	LOW	27.06%	85%
1/5/2026 5:49:29 PM	MEDIUM	52.1%	51.7%

- Stress checker interface used for camera-based stress detection

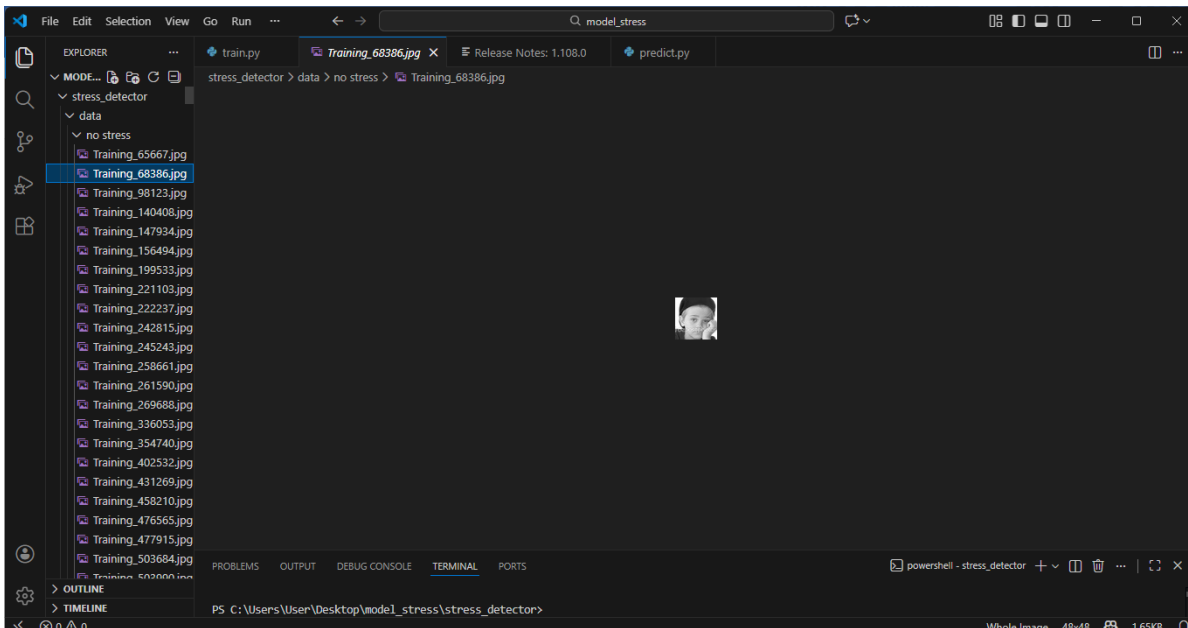


Workload management and assignment screens



These screenshots demonstrate how different user roles interact with the system and validate the usability and functionality of the frontend application.

Appendix B – Stress Detection Dataset



This appendix describes the dataset used to train and evaluate the AI-based stress detection model. The dataset consists of facial images collected and categorized into **stress** and **no-stress** classes based on facial expression characteristics.

The dataset includes variations in facial expressions, lighting conditions, and user appearances to improve model robustness and accuracy. Proper preprocessing techniques such as face detection, normalization,

and resizing are applied before model training. The quality and diversity of this dataset contribute directly to achieving the required stress detection accuracy.

Appendix C – API Endpoint Summary

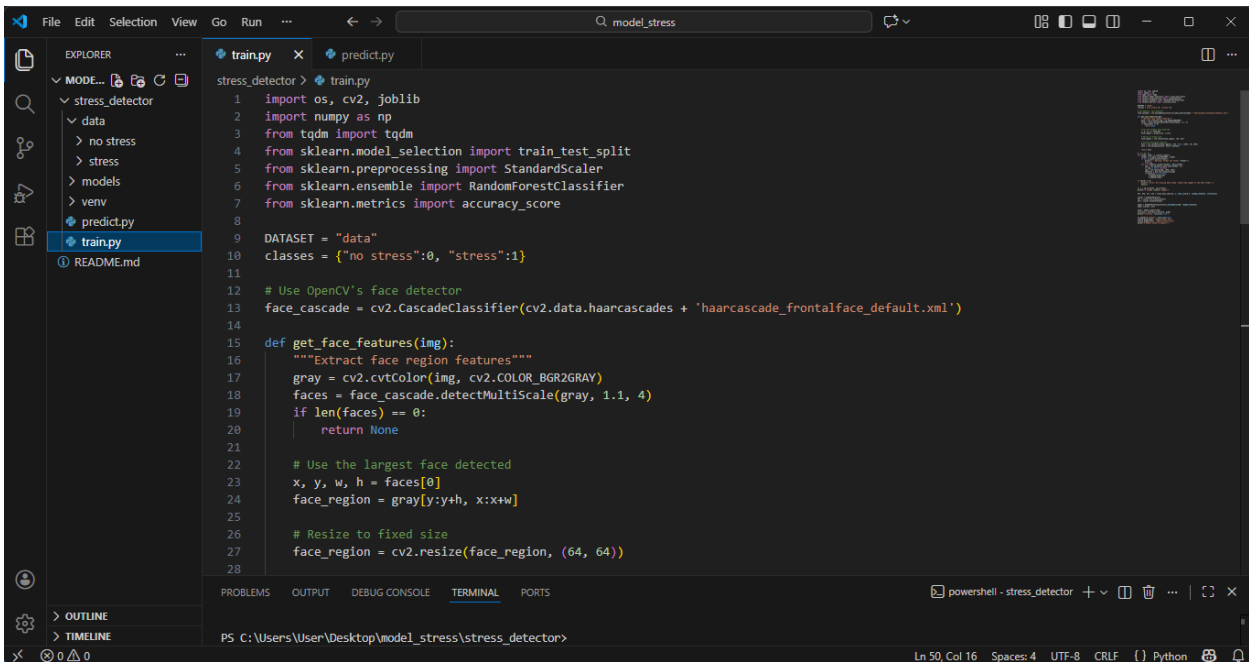
This appendix provides a summary of the key RESTful API endpoints used in the system. These APIs enable communication between the frontend application, backend server, and AI modules.

The API categories include:

- Authentication APIs (login, registration, token validation)
- Stress detection APIs (stress request, image submission, result retrieval)
- Workload management APIs (create, assign, update workloads)
- Notification APIs (real-time alerts and status updates)

Each API endpoint follows RESTful design principles and uses JSON-based request and response formats over secure HTTPS connections.

Appendix D – Stress Classification Logic



```
File Edit Selection View Go Run ...  
EXPLORER  
MODE...  
stress_detector  
data  
no stress  
stress  
models  
venv  
predict.py  
train.py  
README.md  
stress_detector > train.py  
1 import os, cv2, joblib  
2 import numpy as np  
3 from tqdm import tqdm  
4 from sklearn.model_selection import train_test_split  
5 from sklearn.preprocessing import StandardScaler  
6 from sklearn.ensemble import RandomForestClassifier  
7 from sklearn.metrics import accuracy_score  
8  
9 DATASET = "data"  
10 classes = {"no stress":0, "stress":1}  
11  
12 # Use OpenCV's face detector  
13 face_cascade = cv2.CascadeClassifier(cv2.data.haarcascades + 'haarcascade_frontalface_default.xml')  
14  
15 def get_face_features(img):  
16     """Extract face region features"""  
17     gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)  
18     faces = face_cascade.detectMultiScale(gray, 1.1, 4)  
19     if len(faces) == 0:  
20         return None  
21  
22     # Use the largest face detected  
23     x, y, w, h = faces[0]  
24     face_region = gray[y:y+h, x:x+w]  
25  
26     # Resize to fixed size  
27     face_region = cv2.resize(face_region, (64, 64))  
28  
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS  
powershell - stress_detector  
PS C:\Users\User\Desktop\model_stress\stress_detector>  
Ln 50, Col 16 Spaces: 4 UTF-8 CRLF Python
```

This appendix explains the logic used to classify stress levels and map them to workload management decisions. Based on the output of the AI model, stress levels are categorized into **Low**, **Medium**, and **High**.

- **Low Stress:** Employee is considered comfortable; normal workload can be assigned.
- **Medium Stress:** Employee shows moderate stress indicators; workload may be adjusted or monitored closely.
- **High Stress:** Employee is under significant stress; workload is reduced or reassigned to prevent burnout.

This classification logic enables the system to support intelligent, data-driven workload distribution and proactive stress management.